

Computer Graphic Project 2 Report

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1. Lab 6

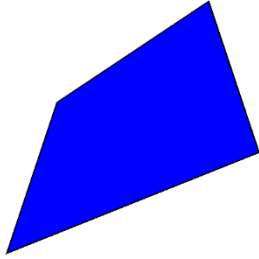


Fig. 1-1: One polygon

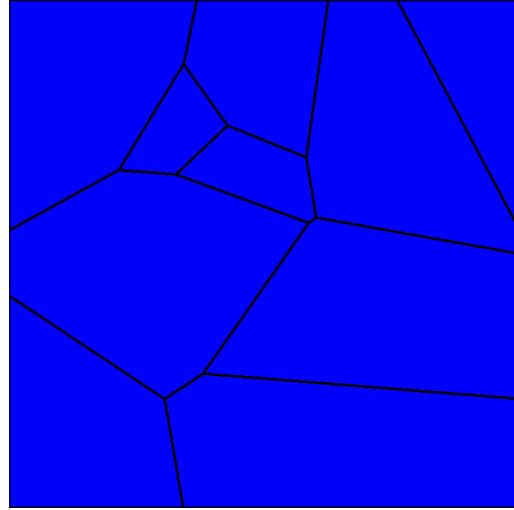


Fig 1-2: Voronoi diagram (N = 10)

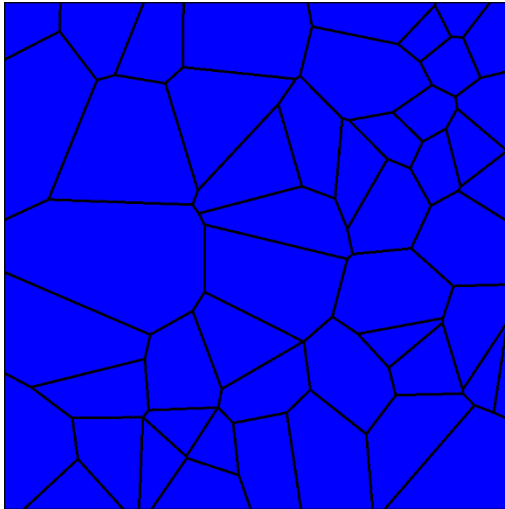


Fig 1-3: Voronoi diagram (N = 50)

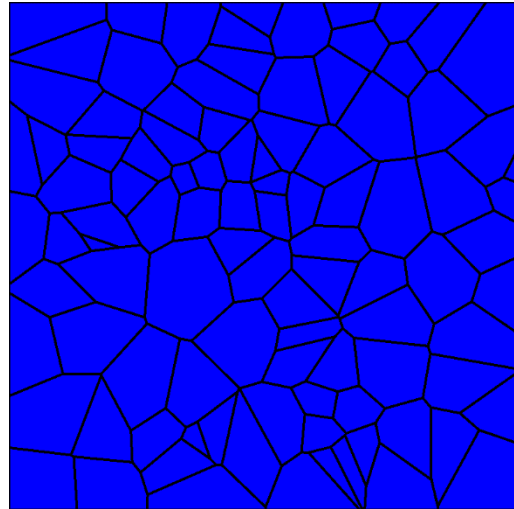


Fig 1-4: Voronoi diagram (N = 100)

In lab 6, we implemented the Voronoi parallel linear enumeration algorithm. We start from a unit square and clip it using the bisector between the current site and the other sites. The final clipped polygon is the Voronoi cell.

Here, a point x belongs to the cell of site P_0 if $\|x - P_0\|^2 \leq \|x - P_i\|^2$. The cell boundaries are defined only by the distances between site, so the areas of the cells vary depending on the spatial distribution of the sites.

Fig. 1-1 is the result of only one polygon. Fig 1-2 to 1-4 show the Voronoi diagrams with different number of random distributed sites N . We can see that as N increase, the cells become smaller and more uniform.

2. Lab 7

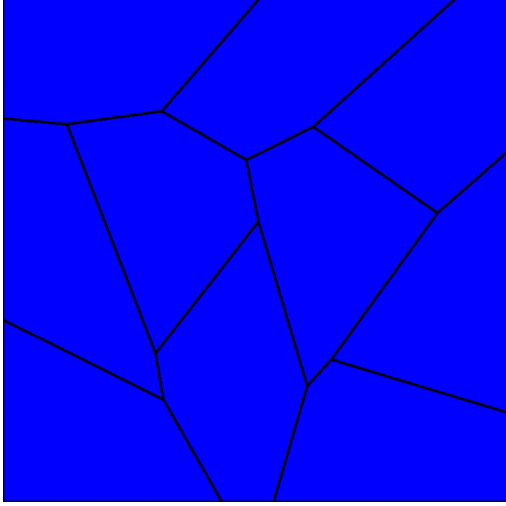


Fig. 2-1: Power diagram ($N = 10$)

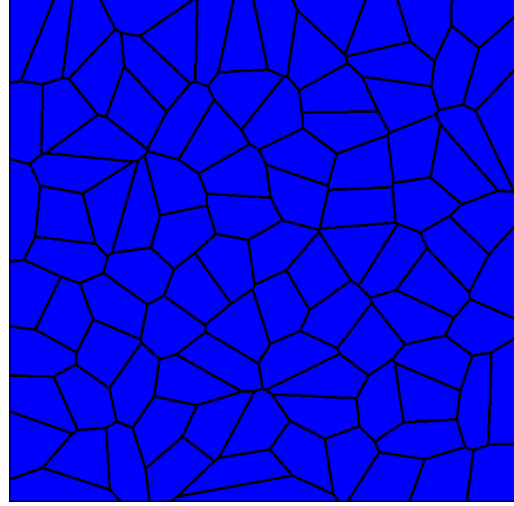


Fig 2-2: Power diagram ($N = 50$)

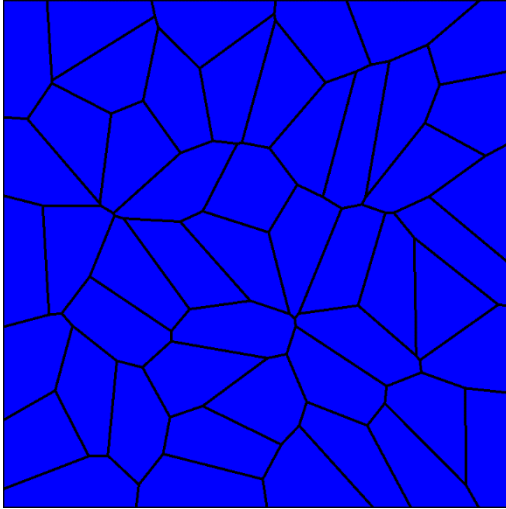


Fig 2-3: Power diagram ($N = 100$)

In lab 7, we implement semi-discrete optimal transport. Now each site P_i has a weight w_i , so to check whether a point x is inside the cell P_0 , we have

$\|x - P_0\|^2 - w_0 \leq \|x - P_i\|^2 - w_i$. In this way, the cell boundaries are shifted

by weight, allowing the area of the cells to be adjusted (target value $\lambda_i = \frac{1}{N}$).

The weights are optimized using LBFGS optimizer. At each iteration, it computes the objective function ($f(x)$) and its gradient ($g[i]$) from the current power diagram. The gradient is defined as the difference between the cell area and the target area ($\lambda_i = \frac{1}{N}$), guiding the optimizer to update the weights until each cell has the same area.

Fig. 2-1 to 2-3 show the power diagram with different N . We can see that the area of each cells are indeed more uniform than in lab 6.

3. Lab 8

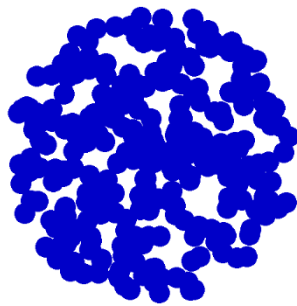


Fig. 3-1: frame 0

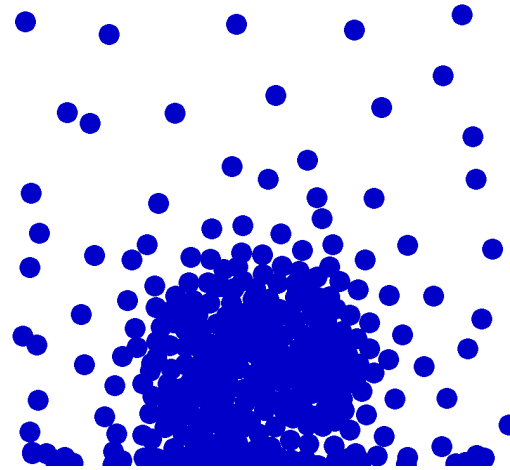


Fig. 3-2: frame 160

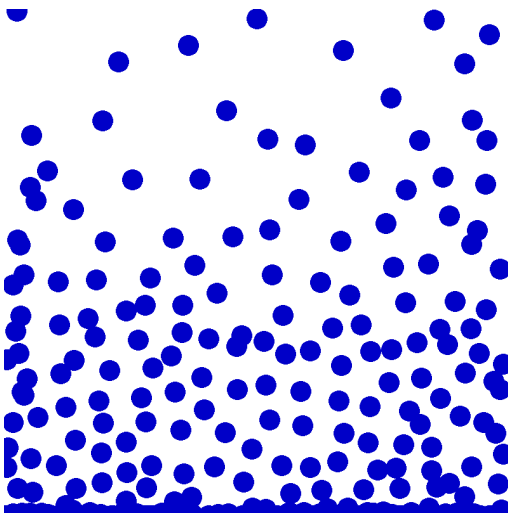


Fig. 3-3: frame 800

In lab 8, we implement the semi-discrete optimal transport fluid simulator with free surfaces. The fluid is represented by 300 particles, and each particle owns a Laguerre cell, each Laguerre cell is clipped by the disk. The cell weight is optimized using the LBFGS optimizer. Then we add a force from the particle position to the cell centroid. The particle's velocity and position can be updated using de Gallouet-Merigot incompressible Euler scheme.

Fig. 3-1 to 3-3 are some frames of the simulation. Fig. 3-1 is the first frame, the particles start from a disk in the middle of the scene. Fig. 3-2 is frame 160, the particles fall to the ground and start to bounce. Fig. 3-3 is frame 800, the particles bounced up. The full simulation video (fluid6.mp4 and fluid6.gif) is on my [GitHub repo](#).

4. Feedback

I like this course. The content is rich and well-structured. The lecture is interesting and the teacher explains very clear. The labs are difficult, but they also give me hands-on experience implementing these algorithms, and give me a deeper understanding of the geometric and mathematical concepts.